Spatiotemporal Remote Sensing Image Fusion Using Multiscale Two-Stream Convolutional Neural Networks

Yuehong Chen, Kaixin Shi, Yong Ge, *Member, IEEE*, and Ya'nan Zhou

***Abstract*— Spatiotemporal remote sensing image fusion (STF) is a promising way to obtain remote sensing data with both fine spatial and temporal resolutions. Gradual and abrupt changes in land surface reflectance images are the main challenges in existing STF methods. Advanced deep learning techniques present powerful ability in learning image-changed information. Therefore, this article proposes a novel spatiotemporal image fusion method using multiscale two-stream convolutional neural networks (STFMCNNs). Multiscale two-stream convolutional neural networks are proposed to capture different sizes of objects in feature learning from a coarse spatial resolution image and two pairs of coarse and fine spatial resolution (FR) images at other dates. Meanwhile, temporal dependence and temporal consistency are explored as complementary information for STFMCNN. Moreover, a local fusion method is developed to characterize local variation by combining two predicted images derived from each stream. Two experiments on different real images are conducted to demonstrate the effectiveness of STFMCNN. Results show that STFMCNN outperformed three existing methods by predicting more accurate FR images with more preserved changed information.**

***Index Terms*— Multiscale feature, spatiotemporal image fusion, two-stream convolutional neural networks.**

I. Introduction

REMOTE sensing data with both fine spatial and temporal resolutions are fundamental sources for monitoring and mapping land surface dynamics, such as land use/land cover change [1]–[4], seasonal variation of vegetation [5], crop yield estimation [6], and climate change [7]. In past decades, remote sensing technologies have developed rapidly and launched a variety of satellite-borne sensors from coarse spatial resolution (CR) to high spatial resolution, such as moderate resolution imaging spectroradiometer (MODIS), Landsat 1-8, Sentinel 1-3, Gaofen 1-7, and Planet. Among these sensors, no single satellite sensor currently provides global observation data with both fine spatial and temporal resolution due to the tradeoff between these resolutions and cloud contamination [8], [9]. As a result, the fusion of different sources of remote sensing data has achieved increasingly popular during the last years [10]–[12].

Various spatiotemporal remote sensing image fusion (STF) methods have proposed to combine temporally sparse but fine spatial resolution (FR) images with temporally dense but CR images to generate synthesized images with both fine spatial and temporal resolutions, which can address the missing data problem and further improve the interpretability of land surface dynamics. Zhu *et al.* [12] summarized existing STF methods in five groups: unmixing-based, weight function-based, Bayesian-based, learning-based, and hybrid. The multisensor multiresolution technique (MMT), proposed by Zhukov *et al.* [13], is perhaps the first unmixing-based STF method and it estimates the value of fine pixels at prediction date by means of the unmixing results of coarse pixels at the prediction date and the available endmembers of fine pixels. Several extensions of MMT are proposed subsequently [10], such as the spatial-temporal data fusion approach [14]. Weight function-based (also referred to as reconstruction-based or filter-based) STF methods estimate the value of fine pixels at the prediction date by the weighted sum of neighboring similar pixels from available FR images [10], [12], [15]. The spatial and temporal adaptive reflectance fusion model (STARFM) and its variants are widely used methods in this group [11], [16], [17]. Bayesian-based STF methods often employ the Bayesian theory to formulate the fusion process into a maximum a posterior problem by maximizing the conditional probability of FR image at the prediction data relative to the available FR and CR images [18]–[20], such as the Bayesian maximum entropy method [20]. Learning-based STF methods leverage machine learning algorithms (e.g., dictionary-pair learning [21] and deep convolutional neural networks [22], [23]) to build the relationships between CR images and available FR images to estimate the values of fine pixels at the prediction date. Hybrid STF methods take advantage of two or more technologies from the aforementioned four groups to improve the performance of synthesized images at a fine scale in space and time [8], [9], [24], [25]. The flexible spatiotemporal data fusion (FSDAF) is one representative of this group through combining unmixing-based and weight function-based techniques [8]. Recently, Li *et al.* [9] have developed an

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advanced hybrid STF method by incorporating subpixel class fraction change information to improve FR image prediction.

There are two key challenges in estimating an FR image at the date (i.e., the prediction date) a CR image was acquired by integrating the CR image into pairs of CR–FR images at other dates in the same area [9], [12]. First, the gradual changes induced by vegetation phenology or growth and seasonal changes in solar elevation may result in the variation of end members in spectral response [9]. Second, the abrupt changes caused by land cover class transition lead to the absence of similar fine pixels in available FR images [12]. For gradual changes, most existing STF methods can handle it and successfully predict pixels associated with vegetation phenology and soil moisture gradual changes [8]. For abrupt changes, it is the most difficult case for STF methods because the abrupt changes can be captured only in the CR image at the prediction date but not in available FR images at other dates [12]. Zhu *et al.* [8] proposed the FSDAF method to downscale the CR image at the prediction date and redistribute the residuals for abrupt change pixels with the help of one pair of CR and FR images at other dates. FSDAF is able to estimate distinct abrupt changes in a CR image at the prediction date but it is hard to reconstruct the tiny abrupt changes in the CR image [12]. The one-pair case STF methods (e.g., FSDAF) use only one pair of CR–FR images at other date and a CR image at the prediction date as inputs to estimate the FR image at the prediction date, resulting in the difficulty in effectively capturing abrupt changes [10], [12], [22]. The CR image at prediction date and two pairs of CR–FR images, one pair acquired before and another after the prediction date (also referred to as two-pair case STF), have been employed to extract abrupt change complementary information from both FR and CR images for improving the reconstruction performance of abrupt changes to estimate the FR image at prediction date [26]–[28]. Compared with methods in the other four groups, learning-based STF methods do not consider gradual and abrupt changes but use a unified framework to establish complex relationships between CR and FR original or difference images [9], [21]–[23]. Recently, deep learning techniques, especially convolutional neural networks, have been demonstrated their strong power in representing images with multiple levels of abstraction and in-feature learning for image classification, semantic segmentation, and change detection [29], [30], and thus, several efforts have been done to exploit deep convolutional neural networks to blend remote sensing images with different resolutions [22], [23], [31]–[33]. These methods often formulate the fusion process as a supervised image super-resolution problem. The image super-resolution convolutional neural network (SRCNN) [34] is widely used and extended in STF [22], [23], [31], [32], [35]. However, SRCNN has only three convolutional layers and it cannot make full use of deep features relative to some state-of-the-art convolutional neural networks [36]–[38]. Meanwhile, effective multiscale features are seldom extracted to capture different sizes of objects to improve STF as the popular multi-scale feature extraction module, atrous spatial pyramid pooling (ASPP), has been demonstrated its efficiency and effectiveness in various applications [37]. Although the convolutional layers with simple different filter sizes are adopted in [22], [23], and [31], they are usually less efficient due to a lot of repetitive computation [37], [39]. Besides, the global fusion parameters derived from each pair of CR images are hard to capture the local variation and they may further lead to the inaccurate prediction of the unknown FR image when two pairs of CR–FR images are used to enhance the prediction by learning-based methods [22].

Therefore, this article proposes a new spatiotemporal remote sensing image fusion method using multiscale two-stream convolutional neural networks (STFMCNNs) for a two-pair case. STFMCNN proposes a deeper model than existing SRCNN-based STF methods [22], [23], [32] and employs ASPP to extract multiscale features for each pair of CR–FR images [37]. Inspired by [22], STFMCNN exploits the temporal dependence (TD) and temporal consistency (TC) of the input FR images and they are used as constrained complementary information to sufficiently incorporate the two pairs of CR–FR images. A local fusion method is developed to fuse the two predicted FR images derived from each pair of CR–FR images.

The remainder of this article is organized as follows. Section II describes the method. Section III shows experimental results and is discussed in Section IV. Section V presents conclusions.

II. Methodology

Let C_{t_1}, C_{t_p}, and C_{t_3} be the input CR remote sensing images at a previous date t_1, prediction date t_p, and posterior date t_3, respectively. We assume the three CR images should be available and used as inputs. The difference image between CR images at arbitrary two dates can be calculated by

$$\begin{aligned} \mathrm{DC}_{t_{p1}} &= C_{t_p} - C_{t_1} \\ \mathrm{DC}_{t_{31}} &= C_{t_3} - C_{t_1} \\ \mathrm{DC}_{t_{3p}} &= C_{t_3} - C_{t_p}. \end{aligned} \tag{1}$$

Let F_{t_1}, F_{t_p}, and F_{t_3} be the corresponding FR remote sensing images at the previous date t_1, prediction date t_p, and posterior date t_3, respectively. F_{t_1} and F_{t_3} should be available and employed as inputs. F_{t_p} is unavailable at the same date as C_{t_p} and it requires to be predicted using two pairs of CR–FR images (i.e., the first pair of C_{t_1} and F_{t_1} and the second pair of C_{t_3} and F_{t_3}) and the C_{t_p}. The difference image between FR images at arbitrary two dates can be calculated by

$$\begin{aligned} \mathrm{DF}_{t_{p1}} &= F_{t_p} - F_{t_1} \\ \mathrm{DF}_{t_{31}} &= F_{t_3} - F_{t_1} \\ \mathrm{DF}_{t_{3p}} &= F_{t_3} - F_{t_p}. \end{aligned} \tag{2}$$

According to (2), the unknown FR image F_{t_p} at the prediction date t_p can be computed by either $F_{t_1}+\mathrm{DF}_{t_{p1}}$ or $F_{t_3}-\mathrm{DF}_{t_{3p}}$. Therefore, the core of the proposed STFMCNN method aims to predict the two FR difference images $\mathrm{DF}_{t_{p1}}$ and $\mathrm{DF}_{t_{3p}}$. One straightforward way to predict the two FR difference images is to model the complex relationships between each pair of CR–FR difference images. However, this way is often less effective due to its uncertainty in the ill-posed inverse problem [21], [22]. An alternative way to model the complex

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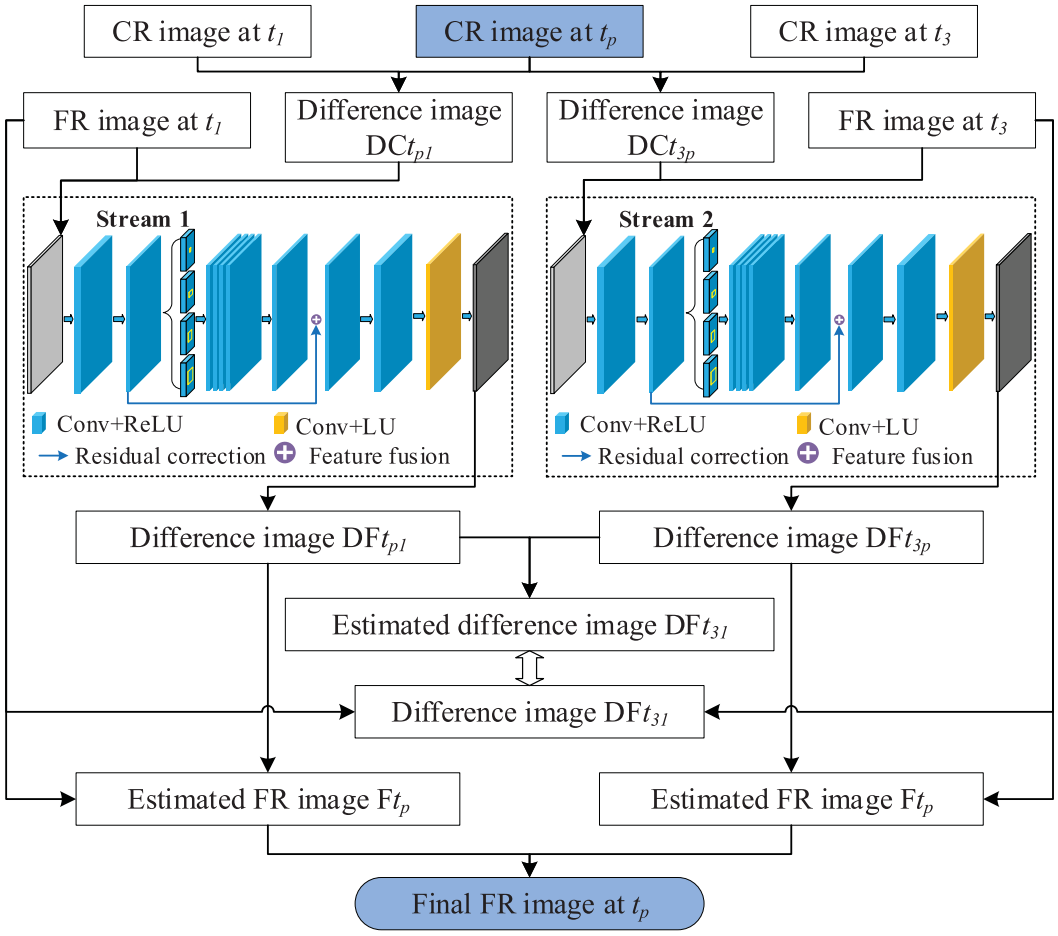


Fig. 1. Flowchart of STFMCNN.

relationships is to combine the CR and FR difference images with the two known FR images F_{t_1} and F_{t_3}, which can be used as a kind of FR complementary information in particular spatial details for the uncertainty reduction in the ill-posed inverse problem [22]. Hence, STFMCNN follows the second way to predict two FR difference images $DF_{t_{p1}}$ and $DF_{t_{3p}}$ from the two pairs of CR–FR images and the CR image at the prediction date. Finally, the two predicted FR difference images are combined with the two known FR images F_{t_1} and F_{t_3} to obtain the unknown FR image at the prediction date t_p. As shown in Fig. 1, STFMCNN performs through five main steps.

1) Preprocessing time series CR images, two FR images at previous and posterior dates, and CR and FR difference images.
2) Designing the multiscale two-stream convolutional neural networks with deeper structure than that of the widely used SRCNN in STF.
3) Training the networks with two FR images and three CR difference images.
4) Predicting the two FR difference images $DF_{t_{p1}}$ and $DF_{t_{3p}}$ with the trained multiscale convolutional neural networks.
5) Fusing the two predicted FR images at date t_p by a local fusion method to yield the final FR image at the prediction date t_p.

Detailed implementation is described as follows.

A. STFMCNN Architecture

Based on the extension of SRCNN [i.e., fast SRCNNs (FSRCNN)] [40], the proposed STFMCNN method employs the depth of FSRCNN with the best performance to generate deeper features than existing SRCNN-based STF methods [22], [23], [31], [32], [35]. Meanwhile, the extensively used ASPP is exploited to extract effective multiscale features for characterizing different sizes of objects in STFMCNN [37]. ASPP uses several atrous convolutional layers with different dilation rates and the same filter parameters, a powerful tool to explicitly adjust the filter's field-of-view, to capture features at multiple scales and its effectiveness has been demonstrated in image semantic segmentation [37], [39]. STFMCNN involves two streams and each one uses the same structure to model the complex relationships between an FR difference image and a CR difference image by integrating an FR image. The first stream of STFMCNN in Fig. 1 takes $DC_{t_{p1}}$ and F_{t_1} as inputs to predict FR difference image $DF_{t_{p1}}$ whereas the second stream uses $DC_{t_{3p}}$ and F_{t_3} as inputs to predict FR difference image $DF_{t_{3p}}$. STFMCNN modifies the structure of FSRCNN [40] and designs a feature extraction convolutional layer from the input layer, a feature shrinking convolutional layer, four atrous convolutional layers with a residual correction to extract multiscale features as the nonlinear mapping, a convolutional layer for feature expansion, and a final convolutional layer for output, as shown in Fig. 1. Except for the final convolutional

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TABLE I
PARAMETERS OF SEQUENTIAL CONVOLUTIONAL LAYERS IN EACH STREAM

	Filter Number	Filter size	Dilation rate
Convolutional layer	128	3×3	-
Convolutional layer	64	3×3	-
Atrous convolutional layer	64	3×3	6
Atrous convolutional layer	64	3×3	12
Atrous convolutional layer	64	3×3	18
Atrous convolutional layer	64	3×3	24
Convolutional layer	64	1×1	-
Convolutional layer	128	3×3	-
Convolutional layer	*B*	3×3	-

layer, each of the other convolutional layers follows the rectified linear unit activation. The final convolutional layer adopts a linear unit activation in order to yield positive and negative output values, which are consistent with the value range in different images. According to [37], the dilation rates (6, 12, 18, and 24) are used in the four atrous convolutional layers and they are concatenated to stacked multiscale features. With the increase of dilation rates, the field-of-view of the atrous convolution filters increases to capture objects in multiple sizes, and thus, the multiscale features can be generated [37], [39]. Subsequently, they are compressed to features by a convolutional layer with the 1×1 filter size for the residual correction, as shown in Fig. 1. The residual correction added to STFMCNN aims to accelerate model learning efficiency and avoid the vanishing/exploding gradients problem for the relatively deeper networks because many previous works have demonstrated its advantages [38], [39]. The detailed parameters of each stream in STFMCNN are presented in Table I. The filter number B in the last convolutional layer means the number of bands in the predicted FR difference image. In STFMCNN, the first stream aims to build the complex relationships between CR difference image $\mathrm{DC}_{t_{p1}}$ and FR difference image $\mathrm{DF}_{t_{p1}}$ by incorporating FR complementary information (i.e., FR image $F_{t_{p1}}$) whereas the second stream intends to build the complex relationships between CR difference image $\mathrm{DC}_{t_{p1}}$ and FR difference image $\mathrm{DF}_{t_{3p}}$ by integrating FR image F_{t_3}. Nevertheless, the FR difference images $\mathrm{DF}_{t_{p1}}$ and $\mathrm{DF}_{t_{3p}}$ fail to achieve due to the absence of FR image at the prediction date t_p. As a result, each stream of STFMCNN cannot conduct the supervised training when lacking the FR difference images $\mathrm{DF}_{t_{p1}}$ and $\mathrm{DF}_{t_{3p}}$ to be the ground truth data in training. Fortunately, according to (2), the difference image $\mathrm{DF}_{t_{31}}$ between FR image at a previous date t_1 and FR image at posterior date t_3 can be estimated by

$$\mathrm{DF}_{t_{31}} = \mathrm{DF}_{t_{p1}} + \mathrm{DF}_{t_{3p}}. \tag{3}$$

Consequently, the outputs of the two streams in STFMCNN can be combined into predicted $\mathrm{DF}_{t_{31}}$ to compare with the known FR difference image $\mathrm{DF}_{t_{31}}$ in training using a loss function (see Section II-B).

If the FR image at prediction date t_p is estimated from each pair of CR–FR images independently, the differences between the two predicted FR images at date t_p may be distinct and they further hamper the prediction of unknown FR images. The combination of the two streams in STFMCNN can maintain the TC of the final FR image at the prediction date t_p estimated from each stream. TC was defined as the consistency between the known FR difference image $\mathrm{DF}_{t_{31}}$ and the combination of the two predicted FR difference images $\mathrm{DF}_{t_{p1}}$ and $\mathrm{DF}_{t_{3p}}$ [see (3)] [22]. In TC, the known $\mathrm{DF}_{t_{31}}$ can be considered as the ground truth, whereas the combination of $\mathrm{DF}_{t_{p1}}$ and $\mathrm{DF}_{t_{3p}}$ can be regarded as the prediction in the supervised learning process. Because the combined two streams in STFMCNN not only model the complex relationships between each pair of CR–FR difference images but also integrate the FR difference image between known FR images F_{t_1} and F_{t_3} to encourage the two predicted FR images from each stream as close as possible. Combined with F_{t_1} and F_{t_3}, the first stream can predict the FR image $F^1_{t_p}$ and the second stream predicts the FR image $F^2_{t_p}$ at the same date, as shown in the following:

$$\begin{aligned} F^1_{t_p} &= F_{t_1} + \mathrm{DF}_{t_{p1}} \\ F^2_{t_p} &= F_{t_3} - \mathrm{DF}_{t_{3p}}. \end{aligned} \tag{4}$$

Under the TC, the two predicted FR images would be as close as possible through minimizing the difference between $F^1_{t_p}$ and $F^2_{t_p}$, which is equivalent to minimize the formula $(F_{t_3} - F_{t_1}) - (\mathrm{DF}_{t_{p1}} + \mathrm{DF}_{t_{3p}})$ in terms of (4). It means to minimize the difference between ground truth $\mathrm{DF}_{t_{31}}$ and estimated $\mathrm{DF}_{t_{31}}$ in (3).

B. STFMCNN Training

As shown in Fig. 1, learning the end-to-end mapping functions $\mathbf{M}_1$ and $\mathbf{M}_2$ for the two streams need the estimation of parameters Θ_1 and Θ_2 in STFMCNN. Under the TC, this can be simply obtained by minimizing the loss between the estimated FR difference image and the corresponding ground truth FR difference image. However, the two known FR images F_1 and F_3 fail to be made full use for enhancing the learning ability of the spatial structure similarity. The TD information can be explored from CR difference image $\mathrm{DC}_{t_{31}}$ to FR difference image $\mathrm{DF}_{t_{31}}$ by incorporating FR image either F_{t_1} or F_{t_3} for this purpose [22]. TD was defined as the correlation of difference image with their neighboring original images in [22]. In other words, the temporal change patterns in difference image are often related to original images, and thus, TD uses an FR image to model its temporal relationship with the predicted FR difference image and improve the mapping ability from a CR difference image to an FR difference image. Therefore, the objective of the loss is designed to minimize the mean-squared error (MSE) between the combined information of TC and TD and the ground truth data, and it is formulated as

$$L(\Theta_1, \Theta_2) = \arg\min_{\Theta_1,\Theta_2} \{L_{\mathrm{TD}} + \alpha L_{\mathrm{TC}}\} \tag{5}$$

where L_{TD} is the MSE of TD information in (6), L_{TC} is MSE of TC information in (6), and α is the parameter

$$\begin{aligned} &L_{\mathrm{TD}}(\Theta_1, \Theta_2) \\ &\quad = \frac{1}{N}\sum_{i=1}^{N} \Big(\big\| \mathbf{M}_1\big(\mathrm{DC}^i_{t_{31}}, F^i_{t_1}; \Theta_1\big) - \mathrm{DF}^i_{t_{31}} \big\|^2 \\ &\qquad + \big\| \mathbf{M}_2\big(\mathrm{DC}^i_{t_{31}}, F^i_{t_3}; \Theta_2\big) - \mathrm{DF}^i_{t_{31}} \big\|^2 \Big) \end{aligned} \tag{6}$$

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where N is the number of training samples

$$L_{\mathrm{TC}}(\Theta_1, \Theta_2) = \frac{1}{N}\sum_{i=1}^{N} \left\| \mathbf{M}_1\left(\mathrm{DC}_{t_{p1}}^{i}, F_{t_1}^{i}; \Theta_1\right) + \mathbf{M}_2\left(\mathrm{DC}_{t_{3p}}^{i}, F_{t_3}^{i}; \Theta_2\right) - \mathrm{DF}_{t_{31}}^{i} \right\|^2. \quad (7)$$

In (5), the known F_{t_1} and $\mathrm{DC}_{t_{31}}$ are employed as the input of one stream to explore the TD between predicted $\mathrm{DF}_{t_{31}}$ and F_{t_1}, whereas the known F_{t_3} and $\mathrm{DC}_{t_{31}}$ are adopted as the input of another stream in STFMCNN to explore the TD between predicted $\mathrm{DF}_{t_{31}}$ and F_{t_3}. The known $\mathrm{DF}_{t_{31}}$ is used as the ground truth data in TD.

In training, the stochastic gradient descent with the standard backpropagation is used to optimize the model [34].

C. STFMCNN Prediction

Using the trained STFMCNN, the FR difference images $\mathrm{DF}_{t_{p1}}$ and $\mathrm{DF}_{t_{3p}}$ can be predicted by

$$\begin{aligned} \mathrm{DF}_{t_{p1}} &= \mathbf{M}_1(\mathrm{DC}_{t_{p1}}, F_{t_1}; \Theta_1) \\ \mathrm{DF}_{t_{3p}} &= \mathbf{M}_2(\mathrm{DC}_{t_{3p}}, F_{t_3}; \Theta_2). \end{aligned} \quad (8)$$

With the two input FR images and two predicted FR difference images, the unknown FR image at the prediction date t_p can be predicted from the first stream in STFMCNN by

$$F_{t_p}^{1} = F_{t_1} + \mathrm{DF}_{t_{p1}}. \quad (9)$$

At the same time, it also can be predicted from the second stream in STFMCNN by

$$F_{t_p}^{2} = F_{t_3} - \mathrm{DF}_{t_{3p}}. \quad (10)$$

D. STFMCNN Fusion

The previous work in [22] used a global fusion weight calculated by an absolute average change of CR images for each predicted FR image, which is hard to capture the local variation for each CR pixel. Therefore, a local fusion method is proposed for each CR pixel to make full use of the advantages of the predicted FR images $F_{t_p}^{1}$ and $F_{t_p}^{2}$ from each stream in STFMCNN. The local fusion method adopts a sliding window for each CR pixel to calculate the local fusion weight. The final FR image F_{t_p} is fused by the proposed local weighting method as

$$F_{t_p}(u) = \omega(v) \cdot F_{t_p}^{1}(u) + [1 - \omega(v)] \cdot F_{t_p}^{2}(u) \quad (11)$$

where $F_{t_p}(u)$ is the final predicted values in FR pixel u, $F_{t_p}^{1}(u)$ is the predicted values in FR pixel u from the first stream of STFMCNN, $F_{t_p}^{2}(u)$ is the predicted values in FR pixel u from the second stream of STFMCNN, and $\omega(v)$ is the local fusion weight from CR pixel v into which the FR pixel u falls and it is computed as

$$\omega(v) = \frac{1/|\mathrm{DC}_{t_{p1}}(v)|}{1/|\mathrm{DC}_{t_{p1}}(v)| + 1/|\mathrm{DC}_{t_{3p}}(v)|} \quad (12)$$

where $\mathrm{DC}_{t_{p1}}(v)$ is the averaged value of all CR pixels within the sliding window centered at pixel v for all bands in CR difference image $\mathrm{DC}_{t_{p1}}$ and $\mathrm{DC}_{t_{3p}}(v)$ is the averaged value of all CR pixels within the sliding window centered at pixel v for all bands in CR difference image $\mathrm{DC}_{t_{3p}}$. Note that all FR pixels falling into CR pixel v use the same weight.

III. Experiments

A. Experimental Design

To evaluate the performance of the proposed STFMCNN method, two experiments on different real images were carried out. The first experiment aims to fuse Landsat-8 images as CR images and Sentinel-2 images as FR images whereas the second experiment intends to blend MODIS images as CR images and Landsat-5 images as FR images. The first experiment has obvious gradual changes and a few abrupt changes, while the second experiment has obvious abrupt changes and a few gradual changes. The spatial resolution ratios of FR and CR images in the first experiment and the second experiment are three and sixteen, respectively. The impact of spatial resolution ratios on the performance of the four STF methods was further investigated by the two experiments because different spatial resolution ratios play an important role in STF [41]. The surface reflectance products (i.e., Level 2 products) of Landsat images were acquired from United States Geological Survey (USGS) (http://earthexplorer.usgs.gov/). The MCD43A4 surface reflectance products of MODIS images were downloaded from USGS as well and MCD43A4 products were daily surface reflectance images adjusted by the bidirectional reflectance distribution function. The surface reflectance products (i.e., L2A products) of Sentinel-2 multispectral images were acquired from European Space Agency (ESA) (https://scihub.copernicus.eu/dhus/#/home). In the first experiment, four spectral bands from both Landsat 8 and Sentinel-2 surface reflectance images were used: blue (B), green (G), red (R), and near infrared (NIR) bands. In the second experiment, six spectral bands from both MCD43A4 and Landsat-5 surface reflectance images were used: Blue (B), Green (G), Red (R), NIR, and two short wavelength infrared bands (SWIR-1 and SWIR-2).

For training the proposed STFMCNN, the three input CR difference images were first resampled with the same size as the two input FR images, and then they were randomly cropped to patches with 128×128 pixels. Five thousand patches were achieved for each experiment. The batch size is set to 8 to fit into the GPU memory and the number of training iterations is set to 50. To optimize STFMCNN in the training phase, the momentum, the weight decay, and the learning rate are set to 0.9, 10^{-6}, and 0.01, respectively. STFMCNN is implemented with Keras package and runs on an NVIDIA GeForce GTX 1080Ti GPU with 11 GB of RAM. The parameter of α in (5) is set to an empirical value of 1 in [22]. Note that the two experiments used the same structure of STFMCNN except the number of input and output image bands. Each experiment should separately perform the training to yield the model for handling different experimental data.

STARFM is a very popular weight function-based STF method and it is widely used as the benchmark in previous works [8], [16], [22]. FSDAF is a relatively new hybrid STF

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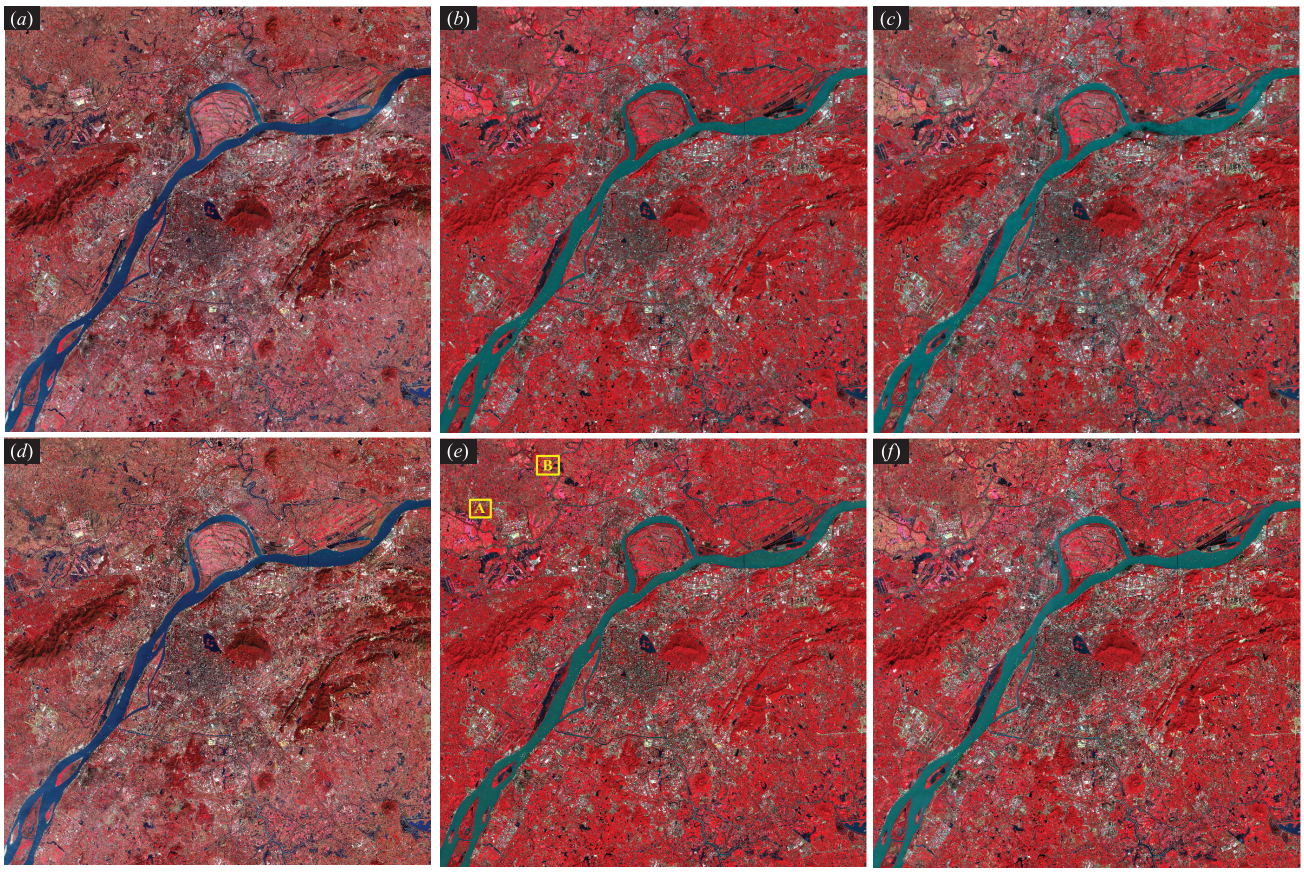


Fig. 2. Experiment on Landsat-8 and Sentinel-2 imagery (standard false-color composition). (a) Landsat-8 image acquired on February 11, 2017. (b) Landsat-8 image acquired on October 9, 2017. (c) Landsat-8 image acquired on October 25, 2017. (d) Sentinel-2 image acquired on February 11, 2017. (e) Sentinel-2 image acquired on October 9, 2017. (f) Sentinel-2 image acquired on October 24, 2017.

method with outstanding performance in reconstructing visible abrupt changes [8], [9], [22]. StfNet is one of the state-of-the-art learning-based STF methods using convolution neural networks [22]. Therefore, FSDAF [8], STARFM [16], and StfNet [22] were compared with the proposed STFMCNN in the two experiments. The default parameters in the three methods were used for a fair comparison. Note that the learning-based method of StfNet employed the same training and validation data sets as STFMCNN in each experiment and the two predicted FR images by FSDAF from each pair of CR–FR images were linearly fused by a global function in StfNet in [22].

Four metrics, the root of MSE (RMSE), structural similarity (SSIM), correlation coefficient (CC), and average difference (AD), were adopted to quantitatively assess the accuracy of predicted FR images [8]. The closer the values of RMSE and AD equal to 0 the more similar the predicted FR image is to the ground truth image. The closer the values of SSIM and CC equal to 1 the more accurate the predicted FR image is. Note that the positive AD means the FR surface reflectance image overestimates, while the negative AD indicates the FR surface reflectance image underestimates.

B. Experiment 1—Landsat-8 and Sentinel-2 Imagery

Three 30-m Landsat-8 surface reflectance images with 2000×2000 pixels and two 10-m Sentinel-2 surface reflectance images with 6000×6000 pixels in Nanjing, China (118°48′13″E, 32°4′33″N) were used to test the four STF methods. The three Landsat-8 images acquired on February 11, 2017 [Fig. 2(a)], October 9, 2017 [Fig. 2(b)], and October 25, 2017 [Fig. 2(c)], are regarded as CR images, respectively. The two Sentinel-2 images acquired on February 11, 2017 [Fig. 2(d)] and October 24, 2017 [Fig. 2(f)] are used as FR images, respectively. Fig. 2(a)–(d) and (f) are used as inputs in training to predict FR image at the same date as the CR image in Fig. 2(b). Fig. 2(e) acquired on October 9, 2017, is the ground truth FR image with the same date as the CR image in Fig. 2(b) to compare with the predicted FR image in accuracy assessment. Fig. 2 shows that there are distinct gradual changes (i.e., phenological changes) between Fig. 2 (a) and the other two images [Fig. 2(b) and (c)] and some changes appear in the top left area.

1) Predicted Sentinel-2 Images: Fig. 3 presents the four target Sentinel-2 images predicted by the four STF methods. Two zoomed-in areas A and B are also displayed in Fig. 3 to highlight the difference between predicted Sentinel-2 images and the ground truth Sentinel-2 image at prediction date t_p and the two input Sentinel-2 images at previous t_1 and posterior date t_3. It can be found from Fig. 3(a)–(d) that the four predicted Sentinel-2 images are very close to the ground truth Sentinel-2 image in Fig. 2(e), suggesting that the four methods can effectively capture the temporal changes

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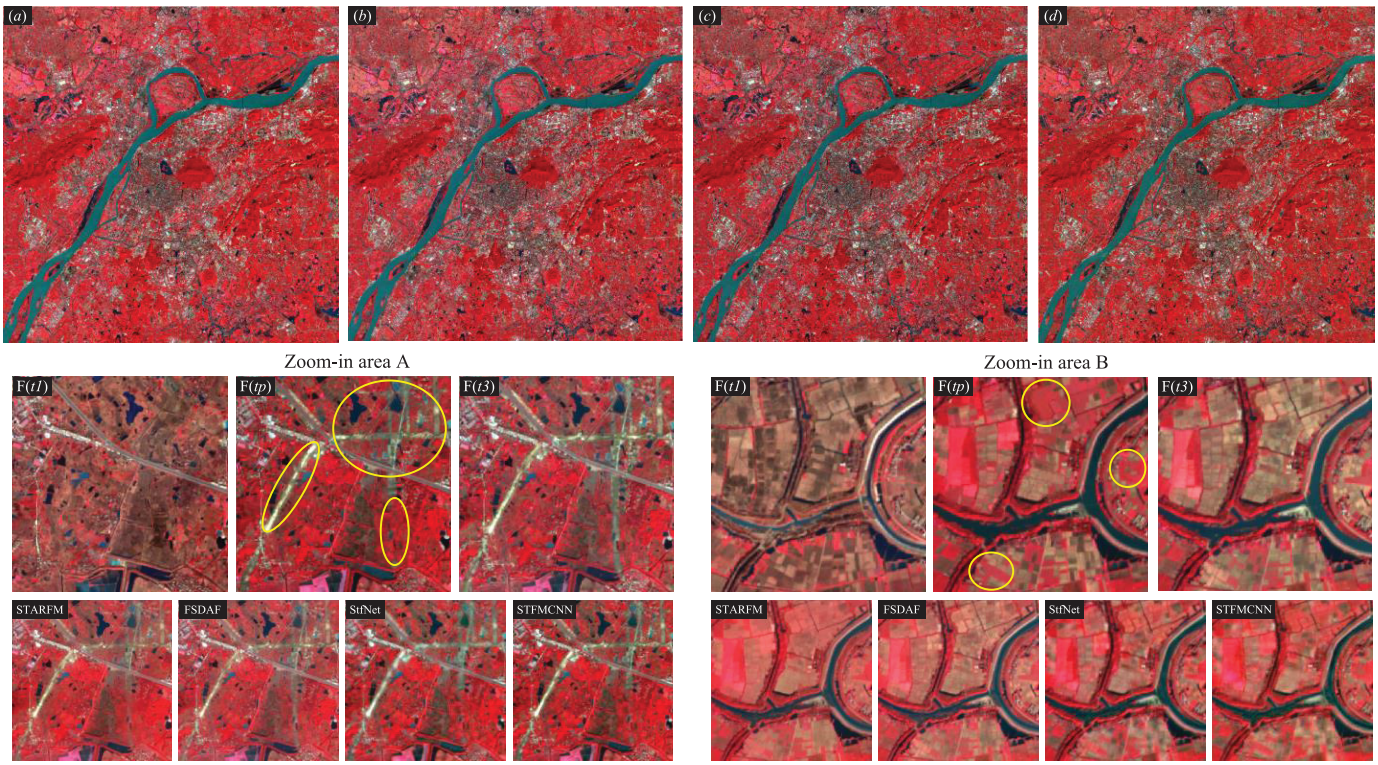


Fig. 3. Predicted Sentinel-2 images, including the zoomed-in area A with abrupt changes and the zoomed-in area B with gradual changes (standard false-color composition). (a) STARFM. (b) FSDAF. (c) StfNet. (d) STFMCNN.

to predict the Sentinel-2 images from the Landsat-8 image by incorporating the two pairs of previous and posterior Landsat-8 and Sentinel-2 images. Fig. 3 displays four STF results and three Sentinel-2 images in the zoomed-in area A with main abrupt changes from farmland to the road and in the zoomed-in area B with main gradual changes by crop phenological variation. In the zoom-in area A of Fig. 3, the abrupt changes at prediction date in the top and left two yellow ellipses were reconstructed by the four methods, while the gradual change in the bottom yellow ellipse at prediction date was not recovered by STARFM and FSDAF but by StfNet and the proposed STFMCNN. FSDAF yielded more wrongly road pixels than STARFM in the bottom yellow ellipse. StfNet and STFMCNN recovered the road in the left ellipse better than STARFM and FSDAF as their color of the road were closer to that in the ground truth image. Especially, STFMCNN recovered fewer blurry roads than StfNet and the best roads in the upper and left two ellipses among the four methods. In the zoomed-in area B of Fig. 3, StfNet and STFMCNN predicted more accurate farmland phenological changes than STARFM and FSDAF, especially in the three yellow ellipses. For the upper yellow ellipse in the zoomed-in area B, the color of the crop by STFMCNN is closer to the ground truth image than the other three results. For the right and bottom two yellow ellipses, the crops were completely recovered by STFMCNN. In contrast, the other three methods recovered only part of them.

2) Accuracy Assessments: Table II shows the accuracy assessment of the four predicted Sentinel-2 images. Comparing the four quantitative metrics calculated using the ground truth Sentinel-2 image in Fig. 2(e) with the four predicted Sentinel-2 images Fig. 3(a)–(d), we can see that the accuracies of both StfNet and STFMCNN are obviously better than those of STARFM and FSDAF. Meanwhile, the proposed STFMCNN generated the best results in terms of the four metrics. Specifically, the average RMSE of STARFM and FSDAF is three times greater than that of StfNet and STFMCNN for all four bands. Comparing StfNet with STFMCNN, the RMSEs of STFMCNN are 77%, 76%, 73%, and 81% of those of StfNet for the four bands, respectively. STFMCNN has the lowest RMSE and the highest CC among the four methods for all bands, suggesting that STFMCNN predicted the closest Sentinel-2 image to ground truth Sentinel-2 image. At the same time, the SSIM of STFMCNN is 0.1181 higher than the average SSIM of the other three methods for all bands, indicating that STFMCNN generated more spatial structure details than the other three methods. The CC of STFMCNN is 0.0825 greater than the average CC of the other three methods for all bands. The ADs of STARFM, FSDAF, and StfNet are positive for all bands, suggesting the three methods produced overestimated Sentinel-2 images. But STFMCNN yielded negative ADs for three bands, indicating it predicted underestimated Sentinel-2 image.

C. Experiment 2—MODIS and Landsat-5 Imagery

Three 30-m Landsat-5 surface reflectance images with 4320 × 4320 pixels acquired on November 26, 2004, December 12, 2004, and December 28, 2004, in Australia (149°46′51″E, 28°45′3″S) were employed as FR images in this experiment. Three MCD43A4 surface reflectance images

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TABLE II
ACCURACIES OF DIFFERENT METHODS FOR REAL LANDSAT-8 AND SENTINEL-2 SURFACE REFLECTANCE IMAGE FUSION

	Band	STARFM	FSDAF	StfNet	STFMCNN
RMSE	*B*	0.0453	0.0405	0.0098	**0.0076**
	G	0.0308	0.0264	0.0101	**0.0076**
	R	0.0269	0.0260	0.0112	**0.0082**
	NIR	0.0567	0.0341	0.0166	**0.0134**
SSIM	*B*	0.7170	0.7116	0.9107	**0.9487**
	G	0.8051	0.7928	0.9127	**0.9524**
	R	0.8322	0.8134	0.9319	**0.9622**
	NIR	0.8474	0.8532	0.9583	**0.9712**
CC	*B*	0.8460	0.8326	0.9151	**0.9507**
	G	0.8486	0.8366	0.9170	**0.9542**
	R	0.8643	0.8478	0.9334	**0.9633**
	NIR	0.8711	0.8594	0.9593	**0.9722**
AD	*B*	0.0440	0.0385	0.0006	-0.0012
	G	0.0283	0.0225	0.0006	-0.0008
	R	0.0226	0.0202	0.0012	-0.0006
	NIR	0.0520	0.0177	0.0011	0.0000

at the same acquisition dates as the Landsat-5 images were first reprojected from sinusoidal projection to the Landsat-5 UTM projection with a spatial resolution of 480-m and then they were cropped to 270 × 270 pixels as CR images in Fig. 4(a)–(c) with the same extent of Landsat-5 images in Fig. 4(d)–(f). Fig. 4(a)–(d) and (f) are used as inputs in training to predict FR image at the same date as the CR image in Fig. 4(b). The Landsat-5 surface reflectance image in Fig. 4(e) with the same date as the CR image in Fig. 4(b) is used as the ground truth FR image for accuracy assessment purposes. It can be observed from Fig. 4 that floods at the prediction date resulted in apparent abrupt changes from land to water in the lower left area.

1) Predicted Landsat-5 Images: Four predicted Landsat-5 images through STARFM, FSDAF, StfNet, and STFMCNN are displayed in Fig. 5(a)–(d), respectively. The zoomed-in area A is also shown in Fig. 5 to visually compare the detailed difference between predicted Landsat-5 images and the ground truth Landsat-5 image. It can be seen from Fig. 5(a)–(d) that the spatial distribution of the four predicted Landsat-5 images is similar to that of the ground truth Landsat-5 image in Fig. 4(e), indicating that the four methods can effectively predict the Landsat-5 image from MCD43A4 image with the help of two pairs of previous and posterior MCD43A4 and Landsat-5 images. Even so, the four methods generated different performances on restoring the abrupt changes by floods. Overall, the four methods did not recover the satisfied abrupt changes (i.e., inundated areas), but the two learning-based methods outperformed STARFM and FSDAF. In the zoomed-in area A of Fig. 5, STARFM and FSDAF only restored a few large-size inundated areas. The reason may be that the tiny inundated areas were not well presented in the CR image in Fig. 4(b). StfNet and STFMCNN reconstructed significantly more inundated areas in the zoomed-in area A than STARFM and FSDAF, especially in the three yellow circles. Comparing StfNet with STFMCNN, STFMCNN generated less blurry inundated areas and more spatial details for rivers than StfNet in the zoomed-in area A from visual examination.

2) Accuracy Assessments: The four accuracy assessment metrics for the four predicted Landsat-5 images are shown in Table III. STFMCNN has the smallest RMSE values, the greatest SSIM, and the highest CC for all six bands among the four methods. The SSIM of STFMCNN is 0.0441 greater than the average SSIM of the other three methods for all bands. The CC of STFMCNN is 0.0404 higher than the average CC of the other three methods for all bands. According to the AD values in Table III, STARFM, FSDAF, and StfNet over-estimated Landsat-5 image at prediction date whereas STFMCNN under-estimated Landsat-5 image.

TABLE III
ACCURACIES OF DIFFERENT METHODS FOR REAL MODIS AND LANDSAT-5 SURFACE REFLECTANCE IMAGE FUSION

	Band	STARFM	FSDAF	StfNet	STFMCNN
RMSE	*B*	0.0114	0.0104	0.0101	**0.0095**
	G	0.0149	0.0148	0.0141	**0.0131**
	R	0.0193	0.0190	0.0185	**0.0169**
	NIR	0.0293	0.0300	0.0276	**0.0251**
	SWIR-1	0.0577	0.0578	0.0419	**0.0404**
	SWIR-2	0.0455	0.0455	0.0383	**0.0361**
SSIM	*B*	0.7828	0.8057	0.8221	**0.8392**
	G	0.7703	0.7715	0.8015	**0.8240**
	R	0.8085	0.8098	0.8300	**0.8485**
	NIR	0.8718	0.8658	0.8845	**0.9076**
	SWIR-1	0.7831	0.7831	0.8795	**0.8926**
	SWIR-2	0.8307	0.8293	0.8712	**0.8863**
CC	*B*	0.7881	0.8109	0.8388	**0.8502**
	G	0.7790	0.7821	0.8274	**0.8417**
	R	0.8120	0.8153	0.8494	**0.8636**
	NIR	0.8885	0.8847	0.9040	**0.9145**
	SWIR-1	0.8139	0.8157	0.8921	**0.8995**
	SWIR-2	0.8379	0.8401	0.8832	**0.8934**
AD	*B*	0.0034	0.0002	0.0022	-0.0006
	G	0.0009	0.0009	0.0032	-0.0005
	R	0.0001	0.0003	0.0041	-0.0007
	NIR	0.0047	0.0053	0.0039	-0.0001
	SWIR-1	0.0376	0.0380	0.0085	-0.0034
	SWIR-2	0.0204	0.0208	0.0075	-0.0031

IV. DISCUSSION

A. Computational Efficiency

The training and prediction time of StfNet and STFMCNN are compared to evaluate their computational efficiency. According to Table IV, the training time of STFMCNN is more than twice that of StfNet and the prediction time of STFMCNN is almost twice as long as StfNet is in the first experiment. The training and prediction time of STFMCNN are also longer than those of StfNet in the second experiment. In general, StfNet is almost twice as fast as STFMCNN because STFMCNN used deeper and more complex convolutional networks. STFMCNN used the network depth of FSRCNN with the best performance [40] and the two experiments demonstrated the structure of STFMCNN was robust to them. Although the network depth of STFMCNN was proven to be effective in the two experiments, the computational efficiency and accuracy of STFMCNN may vary with the change of network depth. Hence, the impact of network depth in STFMCNN should be further analyzed in future usage.

B. Sensitive Analysis of Sliding Window Sizes

The local fusion method using a sliding window was proposed in STFMCNN to explore local variation for fusing

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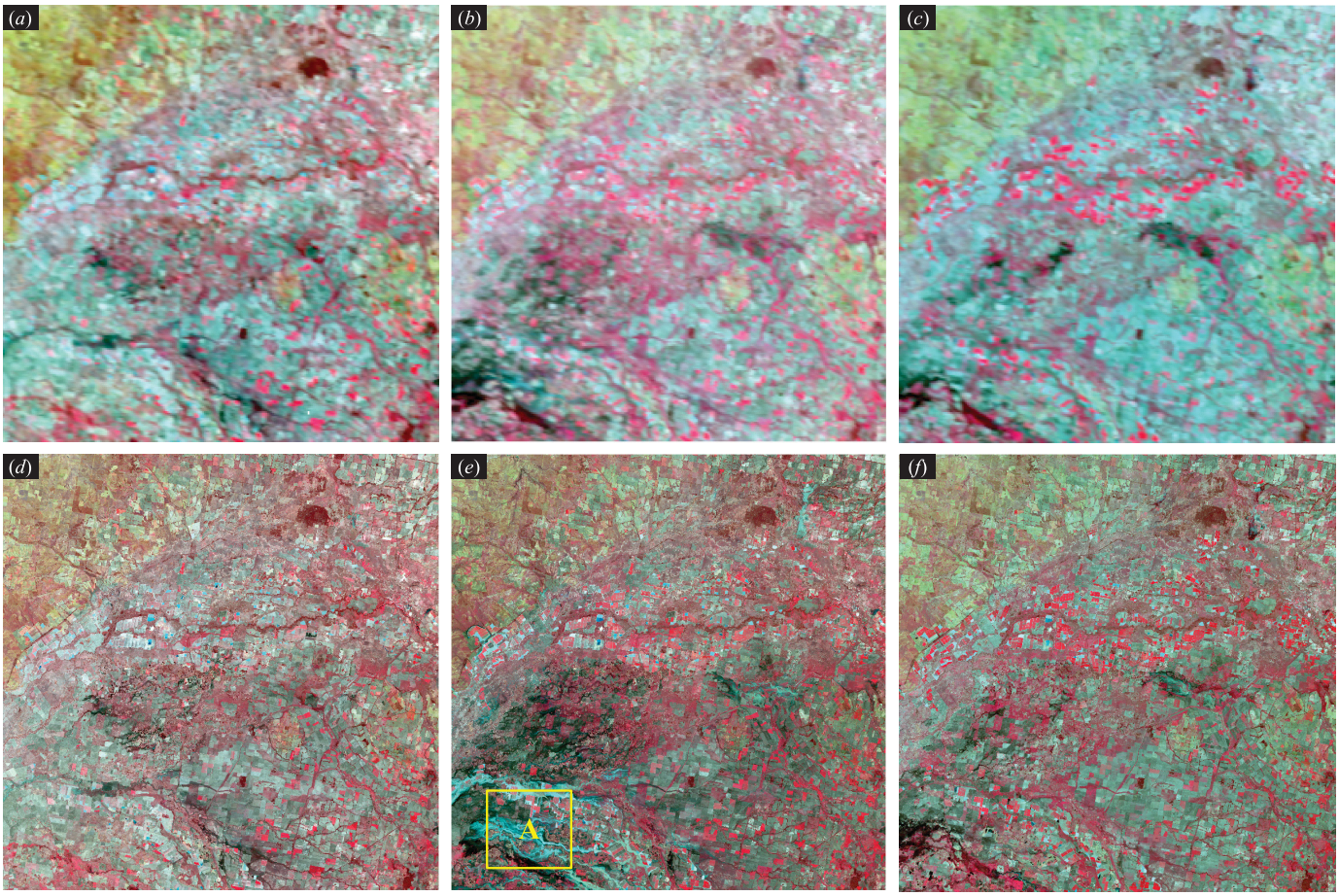


Fig. 4. Experiment on MODIS and Landsat-5 imagery (standard false-color composition). (a) MCD43A4 image acquired on November 26, 2004. (b) MCD43A4 image acquired on December 12, 2004. (c) MCD43A4 image acquired on December 28, 2004. (d) Landsat-5 image acquired on November 26, 2004. (e) Landsat-5 image acquired on December 12, 2004. (f) Landsat-5 image acquired on December 28, 2004.

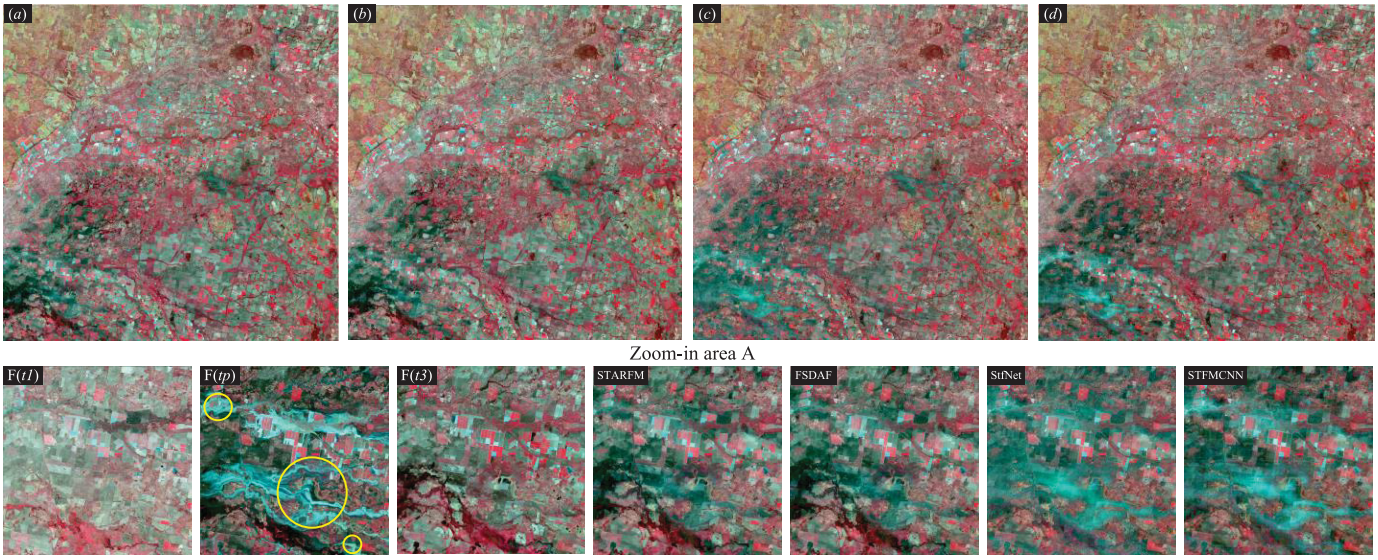


Fig. 5. Predicted Landsat-5 images, including the zoomed-in area A with abrupt changes (standard false-color composition). (a) STARFM. (b) FSDAF. (c) StfNet. (d) STFMCNN.

two predicted FR images. The impact of the size of the sliding window on the performance of STFMCNN is analyzed. Four sliding window sizes (i.e., 1×1, 3×3, 5×5, and 7×7) were adopted in the two experiments and the average RMSE of STFMCNN with different sliding window sizes for each experiment is displayed in Fig. 6. It shows that the best sliding

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TABLE IV
TRAINING AND PREDICTION TIME (S) OF STFNET AND STFMCNN IN THE TWO EXPERIMENTS

	Experiment 1		Experiment 2	
	StfNet	STFMCNN	StfNet	STFMCNN
Training	2156	5413	3708	5467
Prediction	46	81	40	54
Total	2202	5494	3748	5521

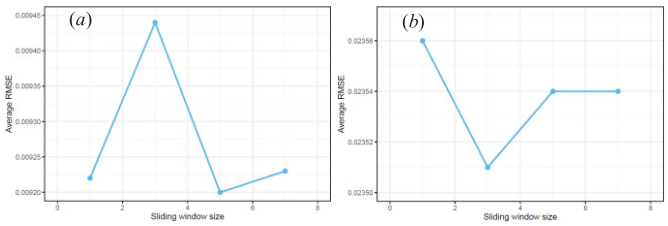


Fig. 6. Average RMSE with different sliding window sizes for STFMCNN. (a) Experiment 1. (b) Experiment 2.

window size is 5×5 for the first experiment and 3×3 for the second experiment, which is used to predict the FR images.

C. Comparison for Global and Local Fusion Methods

STFMCNN developed a local fusion method for capture local variation in blending two predicted FR images, whereas StfNet used a global fusion method. Here, the impact of different fusion methods on the performance of STFMCNN is explored. In the first experiment, the average RMSEs of STFMCNN using global and local fusion methods are 89% and 77% of that of StfNet, respectively, indicating the advanced convolutional networks in STFMCNN decreased 11% RMSE and the local fusion decreased 12% RMSE. In the second experiment, the average RMSEs of STFMCNN using global and local fusion methods are 94% and 93% of that of StfNet, respectively, suggesting the advanced convolutional networks in STFMCNN improved the performance better than the local fusion method. According to the comparison in the two experiments, the advanced neural networks in STFMCNN generally enhance the performance better than the local fusion method. Meanwhile, the local fusion method can improve the performance to a certain degree.

D. Comparison for Two Experiments

Two experiments on different real remote sensing images were tested for the four STF methods and their performances are compared here to analyze the applicability in different scenes. Visual examining the predicted images in Figs. 3 and 5, it can be found that the four methods produced better images in the first experiment than those in the second experiment because slightly more gradual and abrupt changes were restored and less blurry patches were shown in Fig. 3. The accuracy metrics of STFMCNN in Tables II and III also demonstrate that the SSIM and CC values in Table II are larger than corresponding metrics in Table III. Specifically, the SSIM values in the first experiment are over 0.95 for all bands while those values in the second experiment are from 0.82 to 0.91. The CC values in the first experiment are over 0.95, whereas they are from 0.83 to 0.92 in the second experiment for all bands. The average SSIM and CC for the first experiment are 0.0922 and 0.0829 greater than those of the second experiment. In addition, the average RMSE for the first experiment is only 65% of that of the second experiment. The obvious different performances on the two experiments may be caused by three main reasons. First, the spatial resolution ratio of Landsat-8 and Sentinel-2 images is only three while the ratio of MODIS and Landsat-5 images is sixteen, suggesting that the uncertainty of the ill-posed STF problem is significantly higher in the second experiment than that in the first experiment. Second, the gradual changes are the main changes in the first experiment, while the abrupt changes by floods are dominant changes in the second experiment, as shown in Figs. 2 and 4. Third, the spatial resolution of images in the first experiment is higher than that in the second experiment, which provides more abundant spatial context and details that are suitable for convolutional networks. Hence, a smaller spatial ratio between CR and FR fusion images, gradual changes, and more spatial context are beneficial for STF methods to predict more accurate FR images from CR images.

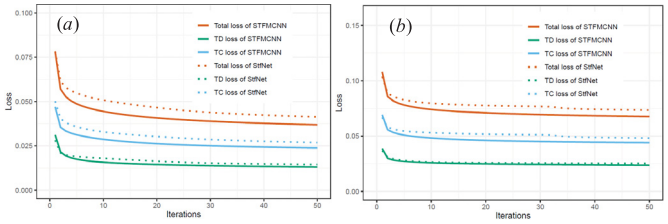


Fig. 7. Loss curves of StfNet and STFMCNN. (a) Experiment 1. (b) Experiment 2.

E. Analysis of TC and TD in Two Learning-based Methods

Both StfNet and STFMCNN explored the TD and TC from two pairs of CR–FR images to add complementary information for the improvement of predicting FR images. Fig. 7(a) and (b) displays the total loss, TD loss, and TC loss curves of StfNet and STFMCNN in the training process for the two experiments, respectively. Fig. 7 shows the loss values of the two methods decrease dramatically at the beginning and gradually converge after 30 iterations in the first experiment and ten iterations in the second experiment. Meanwhile, STFMCNN has lower loss values than those of StfNet for the two experiments, which theoretically indicates STFMCNN performs better than StfNet. In addition, with the use of TC and TD in StfNet and STFMCNN, they generally performed better than STARFM and FSDAF they do not exploit TC and TD information, as shown in Figs. 3 and 5 and Tables II and III.

F. Selection of CR and FR Pairs

STFMCNN requires two pairs of CR–FR images with cloud free and their selection is analyzed for the future usage of the method. In general, similar to previous works [8], [9], [16], [22], each pair of CR–FR images should be acquired at

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the same date or close date with a few abrupt and gradual changes. Otherwise, these changes would be regarded as the changes between the prediction date and other dates, which further reduce the accuracy of FR prediction. According to the two experiments, the first experiment has fewer changes than the second experiment and the prediction performance of the first experiment is better than that of the second experiment. It means that more changes between the prediction date and other dates lead to lower accuracy of the predicted FR images. Therefore, the two pairs of CR–FR images can be chosen closer to the prediction date for accurate FR prediction.

V. Conclusion

This article presents a new spatiotemporal remote sensing image fusion method based on multiscale two-stream convolutional neural networks (STFMCNN) for two pairs of CR–FR images. STFMCNN aims to employ both deeper convolutional neural networks and ASPP to extract deeper multiscale features. Meanwhile, STFMCNN explores the TD and TC information from two pairs of CR–FR images to use as complementary information in the fusion process, which is often ignored in most existing STF methods. Furthermore, a local fusion method is developed to capture local variation in blending the two predicted FR images from each stream. Experimental results have demonstrated the effectiveness of STFMCNN in terms of both visual and quantitative assessments. Hence, STFMCNN is a valuable option for fusing temporally sparse FR images and temporally dense CR images.

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